

SC Ask

Fact-checked Star Citizen Q&A

System documentation: what it is, how it works, configuration and operations

Live URL	https://ask.dayonecitizen.com (private: password gate)
Owner	Doc_Flanigan / Day One Citizen
Status	Live, private since 2026-09-29. Production eval 100% (40/40), 0 invented citations.
Repos	doc-flanigan/sc-ask (web app) doc-flanigan/sc-portfolio (corpus + ingest + nightly jobs)
Design docs	sc-portfolio-docs: superpowers/specs/2026-09-28-sc-ask-design.md (see section 14 "As built"), superpowers/plans/2026-09-28-sc-ask.md
Document date	2026-09-29

This document lists configuration variable NAMES only. Secret values live in .env / .env.local (gitignored), Vercel project settings and GitHub Actions secrets.

1. What SC Ask is

SC Ask is a web chat where anyone can ask a question about Star Citizen, or paste something they heard ("didn't a dev say ships won't be wiped?"), and get a short answer that is **grounded only in official Cloud Imperium sources** and the Day One Citizen claims ledger. Every factual sentence cites a numbered source; the source cards (title, date, link, excerpt) come from the index, never from the model, so the tool cannot invent a URL or a quote.

Each answer that checks a claim carries a verdict:

Verdict	Meaning
Supported	An official source confirms it.
Contradicted	An official source explicitly says otherwise.
Changed over time	Sources disagree across dates; the newest official source wins and both dates are named.
No record found	Nothing in the tracked official sources confirms or denies it. Never presented as "false".
(none)	Plain questions ("what is a Hull-C?") get an answer without a verdict badge.

Ground rules built into the system:

- The chat **never writes to the claims ledger**; ledger changes go through the sc-fact-check workflow.
- No referral links or CTAs anywhere in the tool (credibility).
- One question at a time: no conversational context is sent to the model.
- Off-topic questions get "I only cover Star Citizen" and are neither cached nor counted.

2. How it works

Architecture at a glance

```
sc-portfolio (corpus + ingest)      sc-ask (Next.js 15 on Vercel)
  data/*.jsonl + docs/claims/*.md    /          chat page + Most asked
    | chunk -> passages              /login       password gate
    v                                /api/ask       retrieve + Claude Haiku (streamed)
Upstash Vector (hybrid) <----- /api/popular, /api/feedback
Upstash Redis (cache, stats, spend) <---+
Discord #ask-log <-- webhook (answers, daily summary, alerts)
```

Answering one question

- **Gate checks:** kill switch (ASK_ENABLED), question length (1-500 chars), cache lookup, monthly spend cap, per-IP rate limit (built, off while private).
- **Retrieve:** hybrid (meaning + keyword/BM25) search of the *docs* namespace, top 24, at most 3 passages per document, keep 12; plus a dense search of the *ledger* namespace (top 3) kept when score ≥ 0.73 and placed first. Passages are sorted by date so the model sees change over time. Context capped at ~24k chars.
- **Generate:** one Claude Haiku 4.5 call (temperature 0, max 700 output tokens), streamed. The model writes a 3-line header (VERDICT / TOPIC / QUESTION), a --- separator, then the answer with [n] citations.
- **Guard:** a streaming parser splits header from answer; a citation filter removes any [n] that is not a real passage (even when "[1]" and "[2]" arrive in different chunks). '<' and '>' in questions/passages are neutralized so text cannot break the prompt's delimiters.
- **Record:** spend (from the model's usage; estimated if unavailable), cache entry, Most-asked stats, and a card in #ask-log. One automatic retry if the model fails before any text reached the user.

Source authority used by the model: comm-link (official) > CIG-staff dev post (informal) > video (auto-captions, wording approximate). Dev-post text marked "[Quoting non-CIG user NAME:]" is not attributed to CIG.

3. Data sources (corpus)

Source	File	Size at launch	Refreshed by
Comm-links (RSI)	tools/commlink-corpus/data/corpus.jsonl	6,100 (377 image-only skipped)	GitHub ask-sync (nightly)
CIG-staff dev posts	tools/commlink-corpus/data/devtracker.jsonl (in git)	~1,450	devtracker-archive + ask-sync
Official YouTube transcripts	tools/commlink-corpus/data/youtube.jsonl	~1,840 videos	PC scheduled task (nightly)
Day One Citizen claims ledger	sc-portfolio-docs: claims/*.md	192 claims	GitHub ask-sync (nightly)

- Documents are split into ~320-word passages with 50-word overlap (dev posts under 320 words and ledger claims stay whole). First load: 54,096 passages.
- Ledger claims with status verified / refuted / unverifiable are indexed; superseded claims are skipped.
- YouTube captions are auto-generated (approximate); videos link to the video, not a timestamp.

4. Answer cache and keeping answers fresh

- Every successful, complete, on-topic answer is cached for 30 days. It is findable by every wording it was asked with (vector id <qid>:<wording hash>) and by the model's cleaned-up question (<qid>:c).
- A new question reuses a cached answer only at dense similarity ≥ 0.985 (paraphrase testing showed "Is the Hull-C in the game?" vs "Is the Hull-E in the game?" scores 0.975, so looser thresholds would serve wrong answers). Clicking an item in Most asked always serves the saved answer.
- Not cached: garbled headers, answers cut off at the length limit, off-topic refusals, failed streams.
- **Nightly invalidation** (after each ingest) evicts (1) answers citing any replaced or deleted document, (2) cached questions that a new passage matches at dense similarity ≥ 0.74 , (3) everything after a bulk load (>2000 new passages). Pending work is kept in data/ask-pending-invalidation.json until it succeeds.
- A thumbs-down on an answer evicts it immediately and logs it to #ask-log.

5. Most asked

- Ranked by the number of **different people** who asked (salted SHA-256 of the IP; no IP is stored). Repeat asks and repeated clicks by one person count once.
- Tabs: **Past 30 days** (rolling) and **All time**; entries show "N people"; topic filter chips.
- A question only appears after POPULAR_MIN_ASKERS different people asked it (1 while private, **3 at public launch**). The list always shows the model's cleaned-up question, never raw user text.
- Topics: Ships, Insurance, Wipes & persistence, FPS, Base building, Economy & trading, Pledging & money, Events & Free Fly, Squadron 42, Release dates & roadmap, Other.

6. Automation and monitoring

Job	Where / when	What it does
ask sync	GitHub Actions, sc-portfolio, daily 04:45 UTC (.github/workflows/ask-sync.yml)	Restores/seeds corpus cache, syncs new comm-links, ingests comm + dev + ledger deltas, runs invalidation, posts the daily summary; posts an alert if any step fails.
SC Ask YouTube sync	Windows Task Scheduler on Doc's PC, daily 03:00 (runs when available)	tools/commlink-corpus/ask/yt-nightly.ps1: fetches new transcripts (en-orig captions first), ingests yt, runs invalidation; alerts #ask-log on failure. (YouTube blocks GitHub runners.)
Vercel deploy	On every push to sc-ask main	Automatic production deploy.

Discord #ask-log (webhook, mentions disabled) receives: a card for every new answer (verdict, topic, sources, cost), a daily line (questions, cache hit rate, spend today/month, new passages, sources-current-to date), 80%-of-cap warnings, eviction notices, and job failures. Vercel budget alerts go to #vercel-budget.

7. Configuration variables

7.1 Web app (sc-ask): Vercel production env and local .env.local

Variable	Purpose	Value / default
UPSTASH_VECTOR_REST_URL / _TOKEN	Vector index (docs, ledger, cache namespaces)	secret
UPSTASH_REDIS_REST_URL / _TOKEN	Redis: cache records, stats, spend, rate limits	secret
ANTHROPIC_API_KEY	Claude Haiku key (dedicated sc-ask workspace)	secret
ASK_PASSWORD	Shared password for the private gate	secret
ASK_COOKIE_SECRET	Signs the auth cookie; salts asker IP hashes	secret (random 48 hex)
ASK_LOG_WEBHOOK_URL	Discord #ask-log webhook	secret
ASK_ENABLED	Kill switch: false = "temporarily offline"	true
ASK_MONTHLY_CAP_USD	App-level monthly Haiku spend cap (80% warning, 100% = cache-only)	10
RATE_LIMIT_ENABLED	Per-IP limit on uncached questions (10/hour, 40/day)	false (true at launch)
CACHE_HIT_SCORE	Dense similarity to reuse a cached answer	0.985
LEDGER_MATCH_SCORE	Dense similarity to include a ledger claim	0.73
POPULAR_MIN_ASKERS	Distinct askers before a question shows in Most asked	1 private / 3 public
EVAL_KEY	Header key that lets the eval bypass cache and stats	secret (random)

7.2 Corpus / ingest (sc-portfolio .env on Doc's PC)

Variable	Purpose
UPSTASH_VECTOR_REST_URL / _TOKEN	Same index as the app
UPSTASH_REDIS_REST_URL / _TOKEN	Same Redis as the app (manifest, corpus date)
ASK_LOG_WEBHOOK_URL	Alerts and eviction notices
ANTHROPIC_API_KEY_SC_ASK	Kept for reference/tests; ingest does not call the model
INVALIDATE_SCORE (optional)	Eviction threshold, default 0.74
ASK_CLAIMS_DIR (optional)	Ledger folder, default ../docs/claims

.env format: no leading spaces before names, no ``` lines; Node's parser rejects them.

7.3 GitHub Actions secrets (sc-portfolio)

Secret	Used by
UPSTASH_VECTOR_REST_URL, UPSTASH_VECTOR_REST_TOKEN	ask-sync ingest + invalidation
UPSTASH_REDIS_REST_URL, UPSTASH_REDIS_REST_TOKEN	ask-sync ingest, invalidation, summary
ASK_LOG_WEBHOOK_URL	ask-sync summary and failure alert
SITES_PUSH_TOKEN (existing)	Checks out the private sc-portfolio-docs repo for the ledger

Workflow env (in ask-sync.yml): ASK_MONTHLY_CAP_USD=10, INVALIDATE_SCORE=0.74, ASK_CLAIMS_DIR. The corpus seed lives in the private release *ask-corpus-seed* (corpus.jsonl.gz).

8. Hosted services, costs and caps

Service	Plan	Expected cost	Cap / guard
Anthropic (Claude Haiku 4.5)	Dedicated sc-ask workspace	~\$0.008 per uncached question; eval run ~\$0.30	Console workspace limit \$15/mo; app cap \$10/mo
Upstash Vector (hybrid: openai/text-embedding-3-small + BM25)	Pay-as-you-go	~\$0.10/month; first load ~\$0.20 once	n/a (tiny)
Upstash Redis	Free	\$0	free-tier limits
Vercel	Pro team (existing)	within Pro allowance	Spend Management limit \$20 -> #vercel-budget
GitHub Actions / PC task	existing	\$0	-

9. Data model

Upstash Vector namespaces

Namespace	Contents
docs	Passages from comm-links, dev posts, YouTube. id = <docId>#<n>; metadata docId, sourceType, title, date, url, author, status, ingestedAtMs, excerpt
ledger	One passage per ledger claim (kept separate: Upstash filters AFTER its nearest-neighbour search, so filtering a small subset out of a large namespace returns nothing)
cache	Cached questions: <qid>:<wording hash> and <qid>:c, metadata.qid

Redis keys

Key	Meaning
ingest:manifest	docId -> contentHash:passageCount (incremental ingest)
corpus:currentTo	Newest source date ("Sources current to")
cache:{qid}	Cached answer record (30-day TTL), includes cited docIds
askers:{qid}, qids:recent, top:all, q:{qid}	Most asked: distinct askers, recency, all-time ranking, labels
popular:30d	Cached 30-day ranking (10 min)
asked:{day}, hits:{day}, misses:{day}	Daily counters for the #ask-log summary
spend:{YYYY-MM}, spend:day:{date}, spend:warned:{YYYY-MM}	Spend in microdollars; 80% warning flag
fb:{qid}	Thumbs up/down counts
rl:h, rl:d, rl:login	Rate-limit windows (questions, login attempts)

10. Operations runbook

Task	How
Run locally	cd sc-ask; npm run dev (needs .env.local)
Unit tests	sc-ask: npm test (64) sc-portfolio: npm run test:ask (32)
Quality eval	sc-ask: npm run eval -- https://ask.dayonecitizen.com (40 questions, ~\$0.30; gate >= 90%, 0 invented citations)

Task	How
Turn it off	Vercel env ASK_ENABLED=false, then redeploy (vercel --prod)
Manual ingest	sc-portfolio: npm run ask-ingest [-- --sources comm,dev,yt,ledger] [--dry-run] [--limit N]; then npm run ask-invalidate
Run nightly sync now	gh workflow run "ask sync" (sc-portfolio)
Run YouTube sync now	Start-ScheduledTask -TaskName "SC Ask YouTube sync"; log: tools/commlink-corpus/data/yt-nightly.log
Clear one bad answer	Thumbs-down it in the chat (evicts + logs)
Change a threshold	Set the Vercel env var, redeploy; re-run the eval

Safety rails: ingest fails loudly if a source file is missing and refuses to delete more than 10% of a source type in one run (alerting #ask-log instead); transient Upstash errors retry with backoff; login allows 5 attempts per 15 minutes per IP.

11. Going public: checklist

- After ~2 weeks of private use: set POPULAR_MIN_ASKERS=3 and RATE_LIMIT_ENABLED=true in Vercel.
- Remove the password gate (small code change in middleware) and redeploy.
- Re-run the eval against production and watch #ask-log / spend for the first days.

12. Known limitations

- Coverage is only as good as the corpus: things said only on stream after the last sync, or on non-CIG channels, are not found ("No record found" is not "false").
- YouTube sync depends on Doc's PC being on (it catches up at next login); old speechless/non-English videos have no usable captions.
- Some old comm-links carry wiki re-save dates (e.g. 2026-05-25), which can mislead "newest source wins".
- Minor, low impact: a header-less answer with a --- line early loses its prefix; citation ranges like [1-30] pass through unanchored; a YouTube-only run can advance the sources-current-to date.
- Tabled for v2: Discord bot / public #ask-answers / /ask command, send-to-fact-check button, follow-up context, indexable per-question pages.